



Book I. Prop. XXXIX. Theor.

Equal triangles  and  on the same base () and on the same side of it, are between the same parallels. If  , which joins the vertices of

the triangles, be not $\parallel$  , draw  $\parallel$  (pr. I.31), meeting .

Draw .

$\therefore$  $\parallel$  (const.)

 =  (pr. I.37);

but  =  (hyp.);

$\therefore$  = .

a part equal to the whole, which is absurd.

$\therefore$  $\nparallel$ ; and in the same manner it can be demonstrated that no other line except  is $\parallel$ .

$\therefore$  $\parallel$ .

Q. E. D.